

VIDYA VENKAPPA

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Data Scientist & Software Engineer with 5+ years of experience delivering business impact through data integration, analytics, and scalable system design. Skilled in building robust data pipelines, ensuring data quality, and translating insights through visual storytelling. Proficient in Python, SQL, MongoDB, ML pipelines, and deploying solutions on cloud platforms like AWS.

Experience

Cisco Systems

Software Engineer - II

June 2021-June 2024

- Developed a machine learning pipeline that reduced chatbot query response time by 70% by building a retrieval augmented generation (RAG) pipeline. Managed structured and unstructured data for improved LLM (Large Language Model) responses in the chatbot.
- Ensured responses are polite regardless of the user's query tone, with a 90% positive sentiment score in our custom LLM (Large Language Model) by developing politeness features.
- Expanded the customer base from 1000 to 1100+ (around 100 bots increase) by leading the team in restructuring BotLite (chatbot framework application) from monolithic into a micro-service architecture.

Software Engineer I

Jan 2019-June 2021

- Enhanced bot performance evaluation in BotLite (chatbot framework) by developing an analytics dashboard using Angular's Highcharts and Python libraries, providing visual analytics on user interactions, retention, health status, traffic, and usage.
- Led UI improvements in BotLite (chatbot framework), eliminating redundant features and enhancing user-friendliness, resulting in a spike in bot creations from 400 to 950+ and increase in intent and rule formulations to over 25K+.

Technical Skills

Programming Languages & Tools: Python, SQL, Git, MongoDB, MongoDB Atlas, Snowflake, Looker, Jira, Miro

Data Science Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Hugging Face, SciPy, Matplotlib

ML & Cloud Platforms: Spark, LangChain, LangGraph, Azure Copilot, AWS

Technical Expertise: Machine Learning, Generative AI, Predictive Modeling, Data Pipelines, Statistical Analysis, EDA

Data Science Projects

CyberSentinel [↗](#)

Dec, 2024 - In progress

Agentic cybersecurity chatbot

- Developed an agentic cybersecurity chatbot using FastAPI, LangGraph, and ChromaDB to handle complex security queries with structured reasoning.
- Built modular agent workflows for dynamic multi-step decision-making, including query planning, vector retrieval, LLM-based response generation (Gemini), and validation.
- Enabled accurate, explainable, and scalable responses to cybersecurity issues, streamlining threat analysis and decision support.

CRAB.AI: Critical Review and Analysis for Brilliant Papers [↗](#)

Dec, 2024 - Mar, 2025

AI-driven academic paper review platform

- Developed an AI-driven academic paper review platform using React and FastAPI, integrating LLMs (Gemini, Perplexity, DeepSeek) for automated feedback and insights.
- Implemented role-based access control (RBAC) for students and reviewers, enabling structured feedback, AI-assisted reviews, and manual oversight.
- Reduced review time from 4 months to 1 week, significantly improving submission quality, fairness, and process transparency.

Microsoft Digital Defense Report Analysis [↗](#)

Dec, 2024 - Feb, 2025

Threat intelligence extraction using RAG

- Built a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB and LangChain to analyze Microsoft Digital Defense Reports (2020–2024) and extract threat intelligence context.

- Integrated phi-3.5 to generate detailed insights from retrieved data, surfacing trends in emerging cyber risks and vulnerabilities.
- Delivered actionable findings to support strategic security decisions, improving awareness of evolving threat landscapes.

Spotify Song Popularity Prediction and Recommendation

Oct, 2024 - Dec, 2024

Popularity Prediction using boosting models

- Developed a machine learning pipeline using Spotify's dataset to predict song popularity based on audio features like acousticness and loudness.
- Implemented gradient boosting models for accurate prediction and leveraged K-Means clustering to group similar songs.
- Built a personalized recommendation system that enhanced user experience through tailored music suggestions.

Data Science Research Work

Meeting Summarization - SMACC

July, 2023

A system to streamline multi-modal abstractive and extractive summarization.

- Built a meeting summarization system using LLMs (DialogLM, Longformer) with a novel "Chunking and Chaining" technique to handle long transcripts and generate extractive video trailers for quick visual highlights.

Automatic Subtitle Generation for Videos

March, 2020

IEEE Publications - Conference on Advanced Computing and Communication Systems

- In this project, created a media player capable of generating subtitles for offline videos. Approach involved extracting audio from videos and employing speech recognition engines, including CMU Sphinx and DeepSpeech, to generate these subtitles.

Education

University of Washington, Seattle

Jun 2024 - Jun 2025

Master of Science in Information Systems

In-progress (3.83/4)

Courses: Data Analytics, Data Mining, Data Warehouse, Machine Learning & GenAI.

Indian Institute of Science, Bangalore

Aug 2021 - July 2023

M.Tech in Data Science and Business Analytics

3.87/4

Courses: Introduction to Data Science, Data Engineering at Scale, Practical Machine Learning, Deep Learning, Data Mining, Deep Learning for Visual Analysis.

PES University, Bangalore

Aug 2015 - Aug 2019

B.Tech in Computer Science with Data Science Specialization

9.16/10

Courses: Introduction to Data Sciences, Database Management Systems, Big Data, Artificial Intelligence, Machine Learning, Advanced Machine Learning.

Awards and Recognition

- Won Creative Pitch & Product Award at UW MSIS Demo Day for CRAB.AI
- Cisco Emerging Talent of the Year award at ETC India El Premio 2021 (early-career professional).
- First place in the National Cisco Hackfest, earning a project development grant.

Volunteering & Leadership

- Active member of the LGBTQ+ Cisco Pride ERO Core Team, promoting inclusivity and career growth.
- Created chatbots like the "Community Impact Bot" for environmental awareness and also for Cisco's LGBTQ+ community engagement and awareness.
- Volunteered with CareerVillage.org, Concern India Foundation, and GiftAble, developing materials for special needs students and teaching English in government schools.

Hobbies

Carnatic vocal music, Mandala art and painting, Bollywood dance